

CHONGKAI GAO

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EDUCATION

National University of Singapore , Singapore	01/2024 – present
PhD in School of Computing, Advisor: Prof. Lin Shao	
Tsinghua University , Beijing, China	08/2020 – 06/2023
Master in Department of Automation, Advisor: Prof. Feng Chen, GPA 3.74/4.0, Rank 39/161	
Tsinghua University , Beijing, China	08/2016 – 07/2020
B.Eng. in Department of Automation, GPA 3.62/4.0, Rank 38/158	

INTERNSHIP

Stanford Vision and Learning Lab (SVL)	Stanford University
Visiting Scholar (Supervisor: Prof. Fei-Fei Li)	02/2026 – present
World Model for Robot Manipulation	
Tsinghua Embodied AI Lab (TEA Lab)	Tsinghua University
Research Assistant (Supervisor: Prof. Huazhe Xu)	06/2023 – 01/2024
Equivariant Robot Learning for Manipulation Tasks	
Robot Perception and Learning Lab	University of Texas at Austin
Remote Research Intern (Supervisor: Prof. Yuke Zhu)	05/2022 – 02/2023
Robot Learning, Lifelong Robot Learning Benchmark	
Interaction Lab	University of Southern California
Visiting Scholar (Supervisor: Prof. Maja Mataric)	06/2019 – 08/2019
Non-Verbal Communication Game for Cerebral Palsy	

SELECTED HONORS

• Research Achievement Award, NUS School of Computing.	2025
• Outstanding Paper Award, NeurIPS 2025 Embodied World Models for Decision Making Workshop.	2025
• Best Paper Award, ICRA 2025 Human-Centric Multilateral Teleoperation Workshop.	2025
• Best Paper Award on Robot Manipulation and Locomotion, ICRA 2025.	2025
• Best Robotics Paper Award, CoRL 2024 Workshop on MAPoDeL.	2024
• Best System Paper Nomination, CoRL 2022.	2022
• President's Graduate Fellowship, National University of Singapore	2024
• Excellent Intern, Mech-Mind Robotics	2023
• Excellent Graduates, Tsinghua University	2020
• Philobiblion Scholarship, Tsinghua University	2019
• Scientific and Technological Innovation Award, Tsinghua University	2019
• Third Place Award, Southern California Software Pioneer Hackathon	2019
• Best Team Culture Award, International Aerial Robotics Competition (Asia-Pacific Venue)	2019

TEACHING EXPERIENCE

• Teaching Assistant, Advanced Topics in Robotics, NUS. Lecturer: Prof. David Hsu.	2025 Spring
• Teaching Assistant, Intelligent Robots: Algorithms and Systems, NUS. Lecturer: Prof. Lin Shao.	2024 Fall
• Teaching Assistant, Matrix Analysis and Application, Tsinghua University	2021 Spring & 2022 Fall

SKILLS

Python, C++, ROS, $\LaTeX$, PyTorch, Mujoco, Git

PUBLICATIONS

1. **Chongkai Gao**, Zixuan Liu, Zhenghao Chi, Junshan Huang, Xin Fei, Yiwen Hou, Yuxuan Zhang, Yudi Lin, Zhirui Fang, Zeyu Jiang, Lin Shao, “**VLA-OS: Structuring and Dissecting Planning Representations and Paradigms in Vision-Language-Action Models**”, in *NeurIPS 2025*, **Outstanding Paper Award in NeurIPS 2025 Embodied World Models for Decision Making Workshop**, [paper][website]
2. **Chongkai Gao**, Haozhuo Zhang, Zhixuan Xu, Zhehao Cai, Lin Shao, “**FLIP: Flow-Centric Generative Planning for General-Purpose Manipulation Tasks**”, in *ICLR 2025*, **Oral Presentation in CoRL 2024 LEAP Workshop**, [paper][website]
3. Chenrui Tie*, Yue Chen*, Ruihai Wu, Boxuan Dong, Zeyi Li, **Chongkai Gao**[†], Hao Dong[†], “**ET-SEED: Efficient Trajectory-Level SE(3) Equivariant Diffusion Policy**”, in *ICLR 2025*, [paper][website]
4. Zhixuan Xu*, **Chongkai Gao***, Zixuan Liu*, Gang Yang*, Chenrui Tie, Haozhuo Zheng, Haoyu Zhou, Weikun Peng, Debang Wang, Tianyi Chen, Zhouliang Yu, Lin Shao, “**ManiFoundation Model for General-Purpose Robotic Manipulation of Contact Synthesis with Arbitrary Objects and Robots**”, in *IROS, 2024*, **Oral Presentation**, [paper][website]
5. **Chongkai Gao**, Zhengrong Xue, Shuying Deng, Tianhai Liang, Siqi Yang, Lin Shao, Huazhe Xu, “**RiEMann: Near Real-Time SE(3)-Equivariant Robot Manipulation without Point Cloud Segmentation**”, in *CoRL 2024*, [paper][website]
6. Bo Liu*, Yifeng Zhu*, **Chongkai Gao***, Yihao Feng, Qiang Liu, Peter Stone, Yuke Zhu, “**LIBERO: Benchmarking Knowledge Transfer in Lifelong Robot Learning**”, *NeurIPS Track Datasets and Benchmarks 2023*, [paper][website]
7. **Chongkai Gao**, Zekun Li, Haichuan Gao, Feng Chen, “**Iterative Interactive Modeling for Knotting Plastic Bags**”, in *CoRL 2022*, **Best Systems Paper Nomination**, [paper][website]
8. **Chongkai Gao**, Yizhou Jiang, Feng Chen, “**Transferring Hierarchical Structure with Dual Meta Imitation Learning**”, in *CoRL 2022*, [paper]
9. **Chongkai Gao**, Haichuan Gao, Shangqi Guo, Tianren Zhang, Feng Chen, “**CRIL: Continual Robot Imitation Learning via Generative and Prediction Model**”, in *IROS, 2021*, [paper]
10. Zhenyu Wei*, Zhixuan Xu*, Jingxiang Guo, Yiwen Hou, **Chongkai Gao**, Zhehao Cai, Jiayu Luo, Lin Shao, “ **$\mathcal{D}(\mathcal{R}, \mathcal{O})$ -Grasp: A Unified Representation of Robot and Object Interaction for Cross-Embodiment Dexterous Grasping**”, in *ICRA 2025*, **Best Paper Award on Robot Manipulation and Locomotion**, **Best Robotics Paper in CoRL 2024 MAPoDeL Workshop**, [paper][website]
11. Lixin Xu*, Zixuan Liu*, Zhewei Gui, Jingxiang Guo, Zeyu Jiang, Zhixuan Xu, **Chongkai Gao**, Lin Shao, “**DexSinGrasp: Learning a Unified Policy for Dexterous Object Singulation and Grasping in Cluttered Environments**”, in *Arxiv 2025*, [paper][website]
12. Haonan Chen*, Junxiao Li*, Ruihai Wu, Yiwei Liu, Yiwen Hou, Zhixuan Xu, Jingxiang Guo, **Chongkai Gao**, Zhenyu Wei, Shensi Xu, Jiaqi Huang, Lin Shao, “**MetaFold: Language-Guided Multi-Category Garment Folding Framework via Trajectory Generation and Foundation Model**”, in *IROS 2025*, [paper][website]
13. Jingxiang Guo*, Jiayu Luo*, Zhenyu Wei*, Yiwen Hou, Zhixuan Xu, Xiaoyi Lin, **Chongkai Gao**, Lin Shao, “**TelePreview: A User-Friendly Teleoperation System with Virtual Arm Assistance for Enhanced Effectiveness**”, in *Arxiv 2025*, **Best Paper Award on ICRA 2025 Human-Centric Multilateral Teleoperation Workshop**, [paper][website]
14. Zixuan Chen, **Chongkai Gao**, Lin Shao, Jieqi Shi, Jing Huo, Yang Gao, “**ManiLong-Shot: Interaction-Aware One-Shot Imitation Learning for Long-Horizon Manipulation**”, in *AAAI 2026*, [paper]
15. Lixin Xu*, Zixuan Liu*, Zhewei Gui, Jingxiang Guo, Zeyu Jiang, Zhixuan Xu, **Chongkai Gao**, Lin Shao, “**DexSinGrasp: Learning a Unified Policy for Dexterous Object Singulation and Grasping in Cluttered Environments**”, in *RA-Letters, 2025*, [paper][website]
16. Zixuan Liu*, Ruoyi Qiao*, Chenrui Tie, Xuanwei Liu, Yunfan Lou, **Chongkai Gao**, Zhixuan Xu, Lin Shao, “**Contact Coverage-Guided Exploration for General-Purpose Dexterous Manipulation**”, in *Arxiv, 2026*, [paper][website]